V2 Rail Guard + Prolabs Flex Sheet (0.2mm)

by Aarav Jaisingh, Prolabs Robotics

Build guide for Bangalore FTC teams | goBILDA FTC StarterBot 2026-27 (BIOBUZZ) | 96mm Hogback, 71mm POLLEN

NOT ROBOT TESTED. We have checked this in CAD only. The 0.2mm sheet is our biggest worry (see Known issues). Please test one first and share what you find so we can improve it together.

Download the files

3D files folder (V2 Shooter 3D files):
https://github.com/Aarav-111/FTC-Bangalore-Resources/tree/main/FTC%20precise%20shooting%20mechanism/V2%20Shooter%203D%20files

FTC Shooter V2 Rail Guard.stl
V2 rail guard, print 1
Download

prolabs_plain_sheet_0p2.stl
Prolabs Flex Sheet, print 1
Download

Scan for the 3D files

Scan for all shooter docs

More docs in the FTC precise shooting mechanism folder:

BOM & 3D printing Guide.md
Open

FTC Shooting and Intake Mechanism Theory.pdf
Open

Shooter Assembly Guide for V2.pdf
Open

1. Open the link ▶ 2. Click the file ▶ 3. Click Download raw file (top right) ▶ Saved

To get everything at once, we open the repo page and click **Code > Download ZIP**.

How to print (beginner friendly)

1

Open a slicer

Use any free slicer: **Cura, PrusaSlicer, Bambu Studio or OrcaSlicer**.

2

Import the STL

File > Import, or drag the STL onto the bed.

3

Check scale is 100%

No unit conversion. V2 should read about **129 x 149 x 88mm**. The sheet should read **88 x 104 x 0.2mm**. If it is tiny or huge, the units are wrong.

4

Set the orientation

V2: lay it sideways on its flat side. **Sheet:** flat on the bed.

5

Apply the settings

Use the settings table below for each part.

6

Slice and preview

The sheet must show **exactly 1 layer** with all holes visible. V2 should have no support.

7

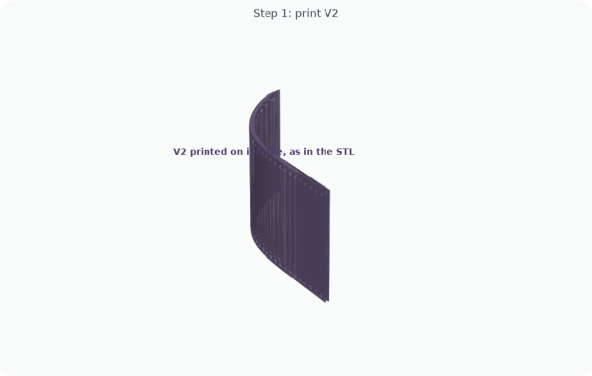
Print and remove

Watch the first layer. Peel the sheet off a flexible plate by bending the plate.

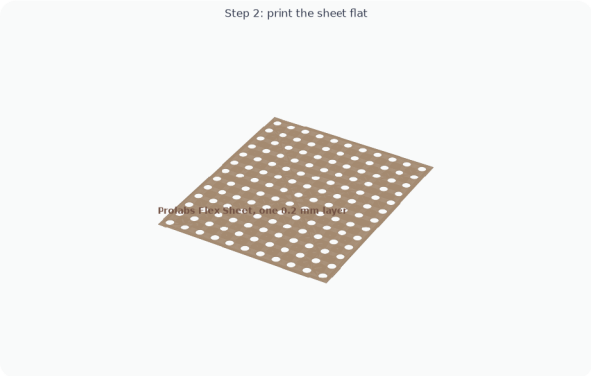
8 Check dimensions

Use calipers. Each size within **+-0.5mm**.

V2 orientation (sideways)



Sheet orientation (flat, one layer)



Setting	V2 rail guard	Prolabs Flex Sheet
Material	PETG or PLA+	PETG
Infill	**100%**	**100%**
Supports	None	None
Layer height	Normal (0.2mm is fine)	**0.2mm, so it is one layer**
Adhesion	5mm brim	No brim; clean, level bed
First layer speed	Normal	**Slow, about 15 to 20 mm/s**
Removal	Normal	Flexible plate, peel gently

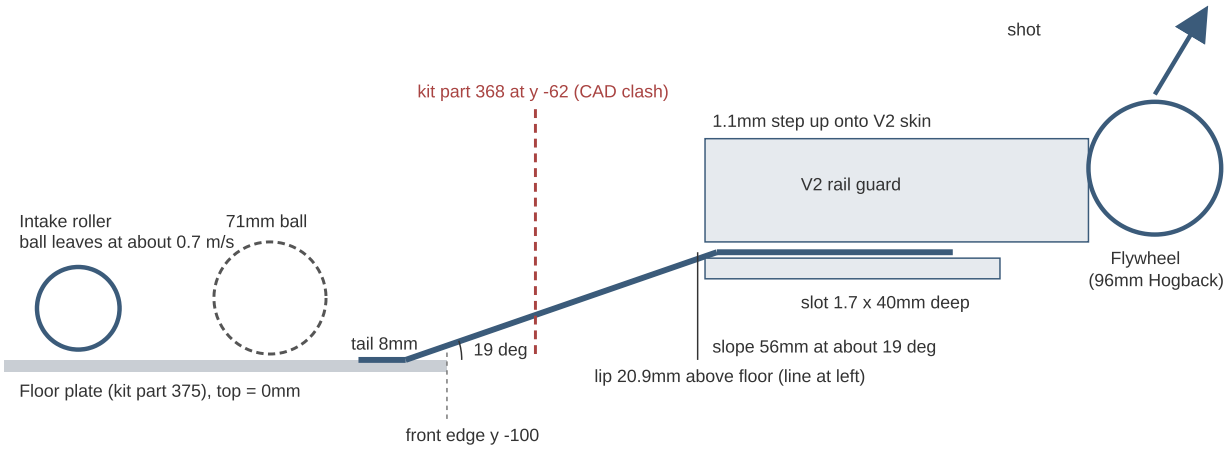
Common problems

Problem	Fix
Sheet will not stick	Clean the bed (soap and water or IPA), re-level, slow the first layer, use a glue stick
Holes closed up	Lower first-layer squish (raise Z offset a little), reduce first-layer flow
Warping (V2)	Keep the brim, raise bed temperature, avoid drafts
Stringing	Dry the PETG, increase retraction, lower nozzle temperature by 5 to 10 C

1. How it shoots (CAD analysis)

This is an analysis from CAD and the STL files. It is not robot tested. Speeds and clearances below are estimates. Test with one ball by hand first.

Side section of the ball path (schematic, not to scale)



Unpowered zone: about 131mm from intake roller to the V2 squeeze. Nothing drives the ball here.

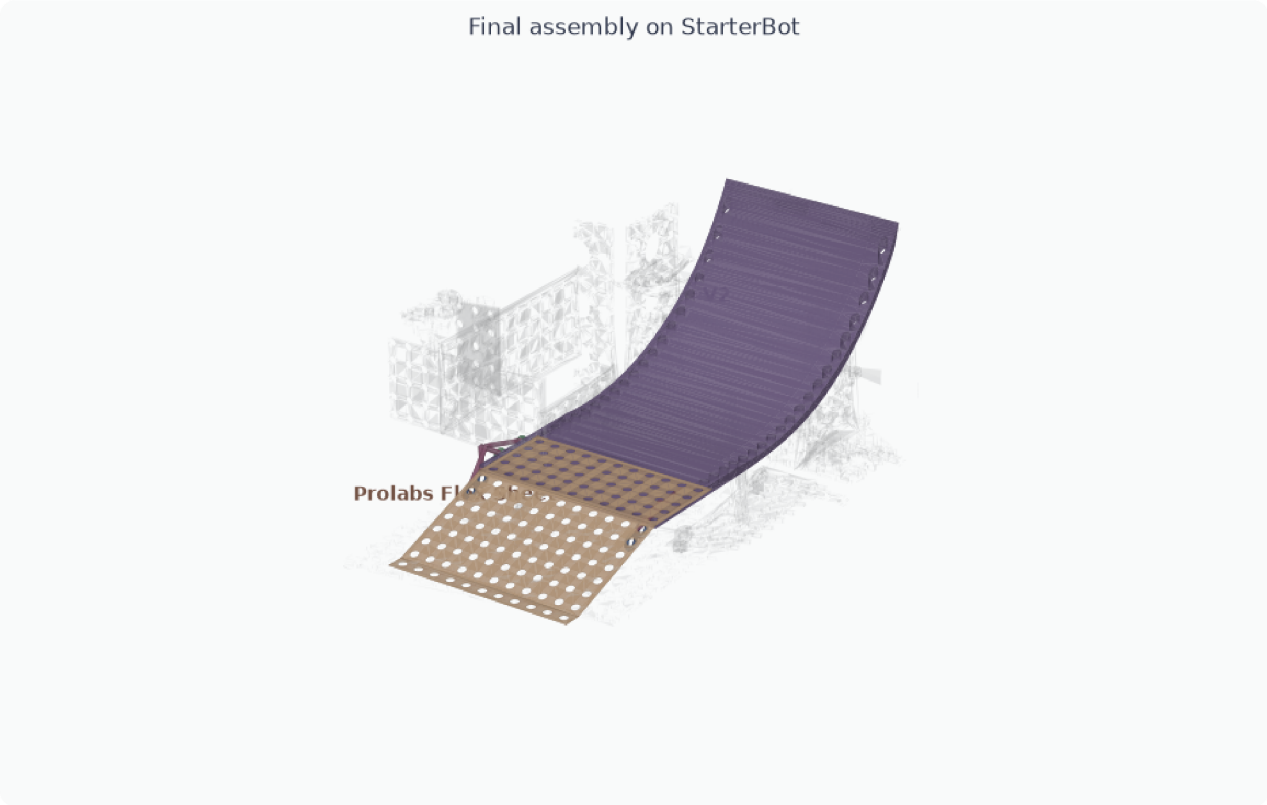
Stage	What happens	Key number
1. Intake roller	Roller throws the ball forward across the floor plate	Needs about 0.7 m/s at exit
2. Floor plate	Ball rolls to the front edge	Front edge y -100
3. Sheet tail and slope	Ball rolls onto the 8mm tail and climbs the 56mm slope	About 19 deg, rise about 18mm
4. Tongue and V2 lip	Sheet tongue sits in the V2 slot. Ball steps up onto the V2 skin	Lip 20.9mm, step 1.1mm
5. V2 rail guard	Rail guard holds the ball against the flywheel	Squeeze peaks at 14.5% on 71mm ball
6. Flywheel and shot	Hogback wheel grabs the ball and launches it along the shooter hood	Not measured

Total unpowered run: about **131mm**, partly uphill. If the ball is slower than about 0.7 m/s it can stop on the slope. A second ball behind it can push it through.

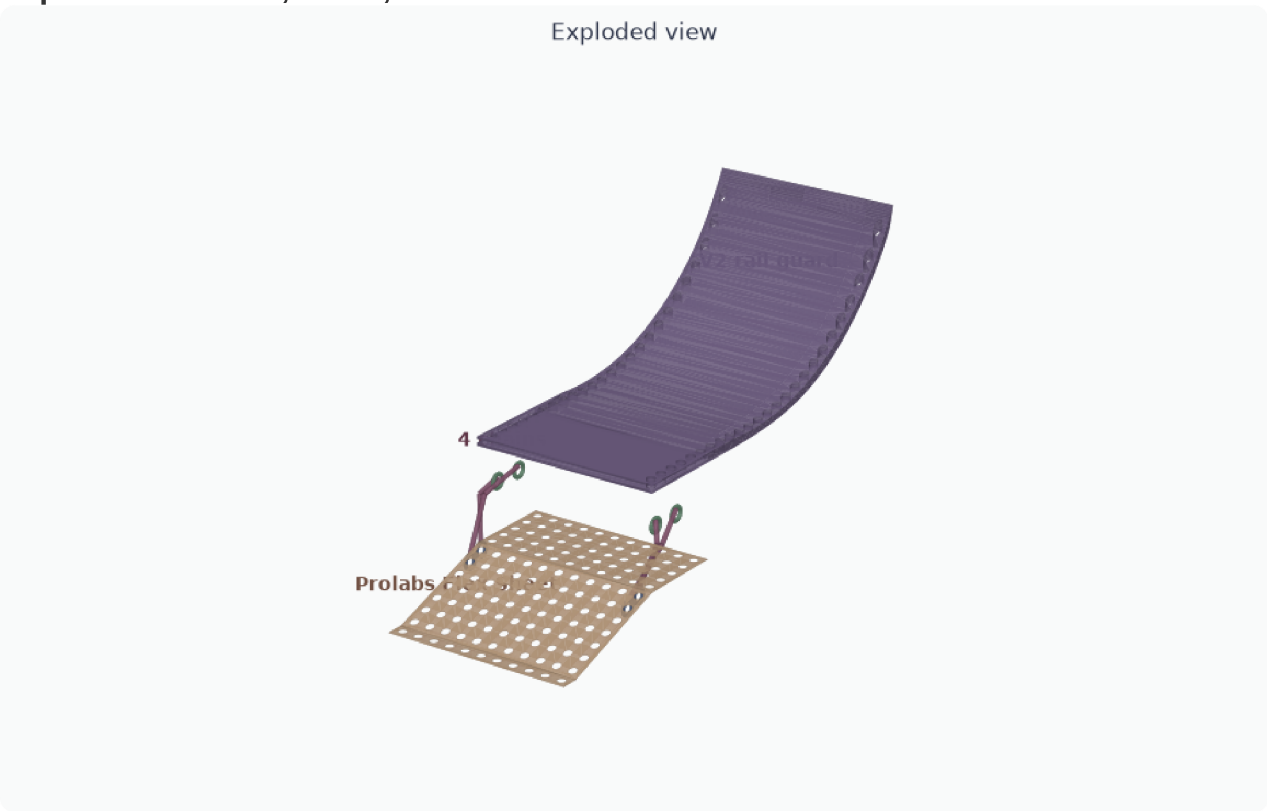
Expected failure points

Where	What can go wrong	What to check
Slope	Ball stalls, too slow from the roller	Roll one ball at match speed
0.2mm sheet	Sags under the ball, bend angle changes	Watch the sheet while rolling
V2 lip	Ball catches on the 1.1mm step	Chamfer the lip, roll by hand
Part 368 (y -62)	Upright plate blocks the path in CAD	Check clearance on the robot
Chains	Band pulls the tongue out or tears holes	Tongue stays 40mm in the slot
Squeeze	14.5% is the top of the 10 to 15% target, can jam soft balls	Feel the ball pass under V2

Final assembly (V2 rail guard and Prolabs Flex Sheet)



Exploded view: V2, sheet, 4 chains



2. Dimensions to double-check

Item	Value	Measure with	Tolerance	Checked
Sheet size	88 x 104mm	Calipers	+-0.5mm	STL
Sheet thickness	0.2mm (one layer)	Calipers / micrometer	+-0.05mm	STL
Hole size	4.0mm	Calipers inside jaws	+-0.2mm	STL
Hole pitch	8.0mm, centres at 4 + 8i	Calipers, over 10 holes = 80mm	+-0.5mm over 10	STL
Hole count	11 x 13 = 143	Count	exact	STL
Edge margin	4mm to hole centre, 2mm to hole edge	Calipers	+-0.3mm	STL
Bend 1 location	40mm (5 holes) from tongue end	Ruler	+-1mm	Design
Bend 2 location	8mm (1 hole) from tail end	Ruler	+-1mm	Design
Bend angle	about 19 deg	Protractor	+-3 deg	Design
Sheet weight	about 1.9g PETG	Scale	+-0.3g	STL volume 1473 mm3
V2 overall size	129.35 x 148.57 x 88mm	Calipers	+-0.5mm	STL
V2 volume / weight	55.0 cm3, about 70g PETG	Scale	+-5g	STL
V2 slot	1.7mm high (z 7.75 to 9.45), 40mm deep	Feeler gauge / 0.2mm sheet test fit	+-0.2mm	STL
V2 side holes	4mm, at x +-40	Calipers	+-0.2mm	STL
V2 hole positions under slot	y -32.5, -25.5, -18.4, -11.3, -4.2 (more behind at 2.9, 10.0 ...)	Calipers from front face	+-0.3mm	STL
V2 hole pitch vs sheet	about 7.07mm vs 8.0mm. Mismatch 0.93mm per hole	Calipers	-	STL

Lip height above floor plate	20.9mm	Calipers depth rod	+-0.5mm	Earlier kit CAD, not re-checked
Step at lip	1.1mm	Calipers	+-0.2mm	Earlier kit CAD, not re-checked
Floor plate front edge	y -100	-	-	Earlier kit CAD, not re-checked
Part 368 position	y -62	Ruler on robot	-	Earlier kit CAD, not re-checked
Squeeze	14.5% peak on 71mm ball	Feel / gap gauge	-	Earlier analysis, not re-checked
Zip ties	small, 2.5mm wide x 100mm (fits 4mm holes)	-	-	Suggested size, not in CAD
Rubber bands	light, about 1.5mm wide x 40 to 50mm long	-	-	Suggested size, not in CAD

Total sheet length check: tongue 40 + slope 56 + tail 8 = **104mm**.

3. Parts list

Item	Qty	Notes
V2 rail guard (printed)	1	rail_guard_v2.stl
Prolabs Flex Sheet (printed)	1	prolabs_plain_sheet_0p2.stl
Zip ties	**8**	2 per chain x 4 chains
Rubber bands	**4**	all the same size
Super glue (thin CA)	1 tube	Used in step 6: zip tie locks (8) and the tongue top edge in the V2 slot. Plus CA primer/activator for PETG
File or knife	-	Chamfer V2 lip
Calipers, 71mm ball	-	Hand checks

4. Print settings

V2 rail guard

- PETG
- On its side
- No supports
- **100% infill**
- 5mm brim

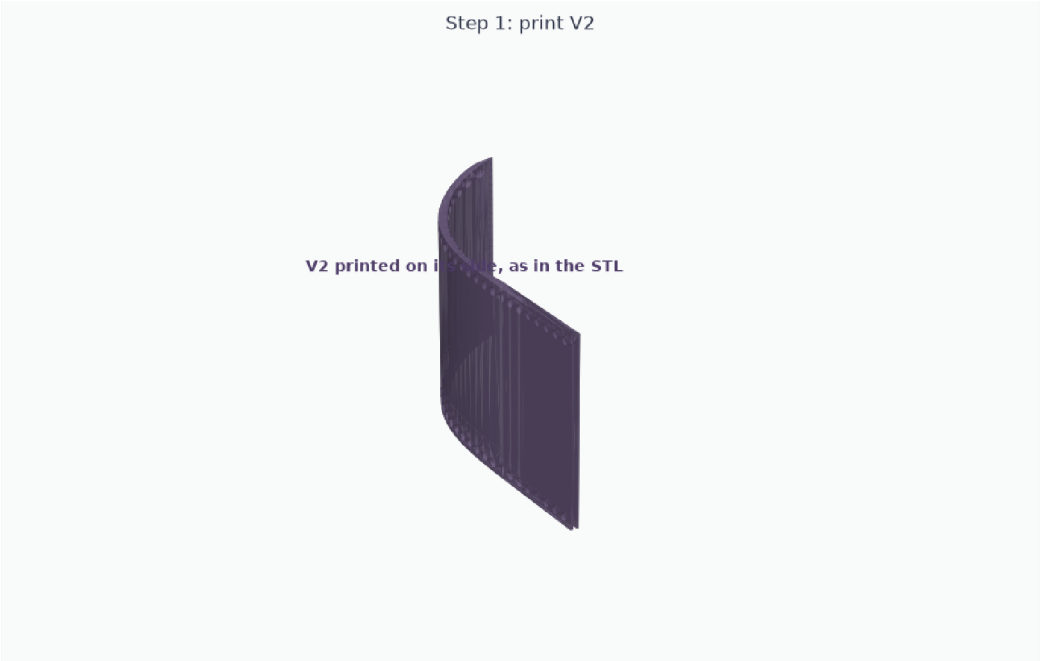
Prolabs Flex Sheet

- PETG, printed flat
- **100% infill**, one 0.2mm layer
- Level the bed carefully
- Slow first layer
- Flexible build plate for removal

5. Step by step

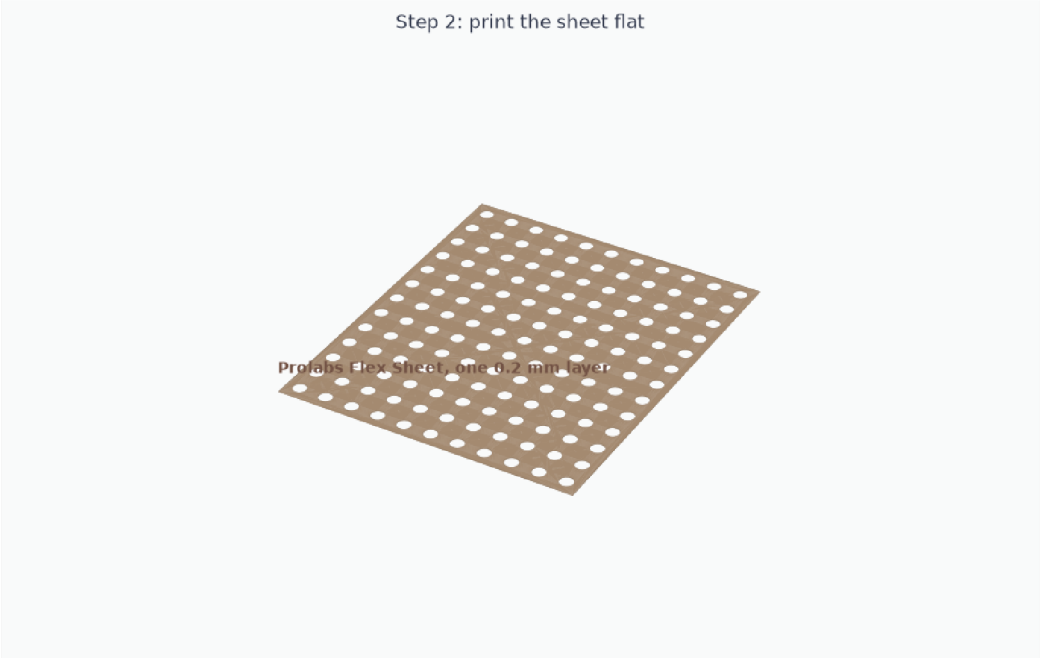
1 Print the V2 rail guard

Print **rail_guard_v2.stl** on its side. **PETG, 100% infill, no supports, 5mm brim.**



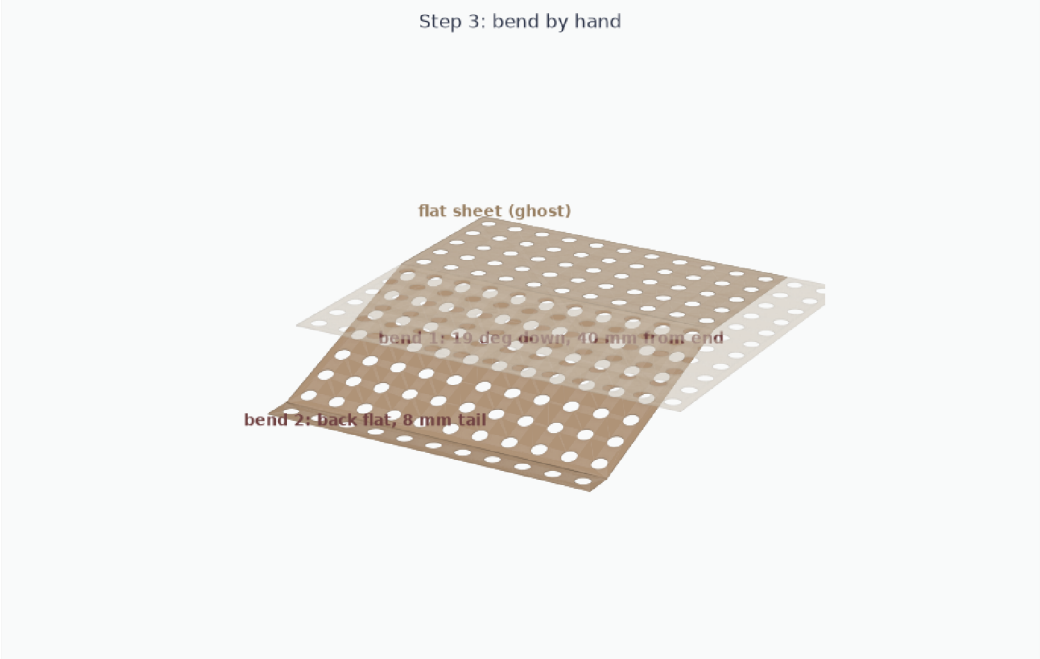
2 Print the Prolabs Flex Sheet

Print **prolabs_plain_sheet_0p2.stl** flat. It is **one 0.2mm layer**, 88 x 104mm, 143 holes of 4mm on an 8mm grid. About 1.9g.



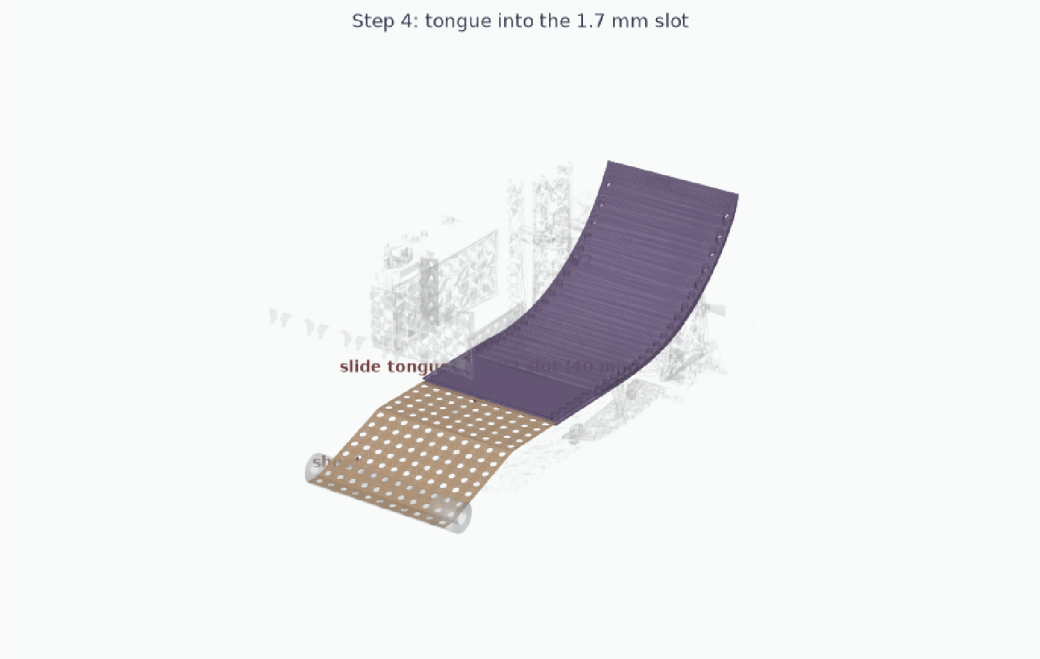
3 Bend it by hand

Bend 1: 19 deg down, 40mm (5 holes) from the tongue end. **Bend 2:** back to flat, 8mm (1 hole) from the tail end. Bend along the webs between hole rows. The sheet is thin, so it bends easily but will not hold the shape on its own.



4 Slide the tongue into the V2 slot

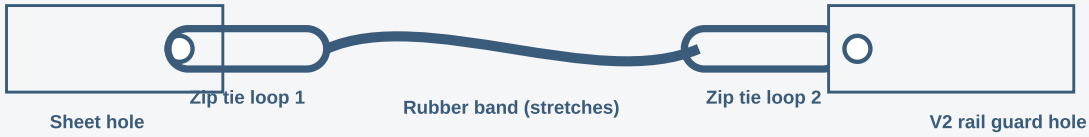
Push the 40mm tongue fully into V2's 1.7mm front slot. The tail runs down to rest on the floor plate.



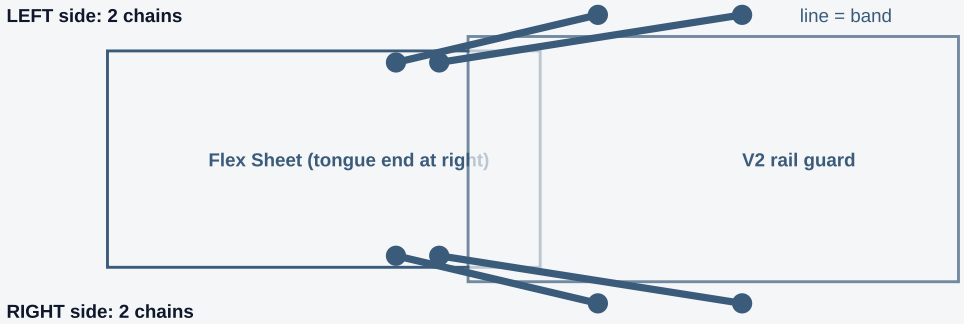
5 Fit the rubber-band chains, both sides

On **each side** make **2 chains**, 4 in total: **sheet hole → zip tie loop → rubber band → zip tie loop → V2 hole**. The bands keep the tongue pulled into the slot.

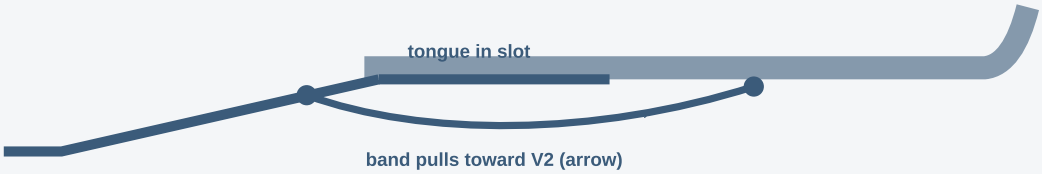
Elastic chain, top view (one chain)



Top view, whole plate: 2 chains per side

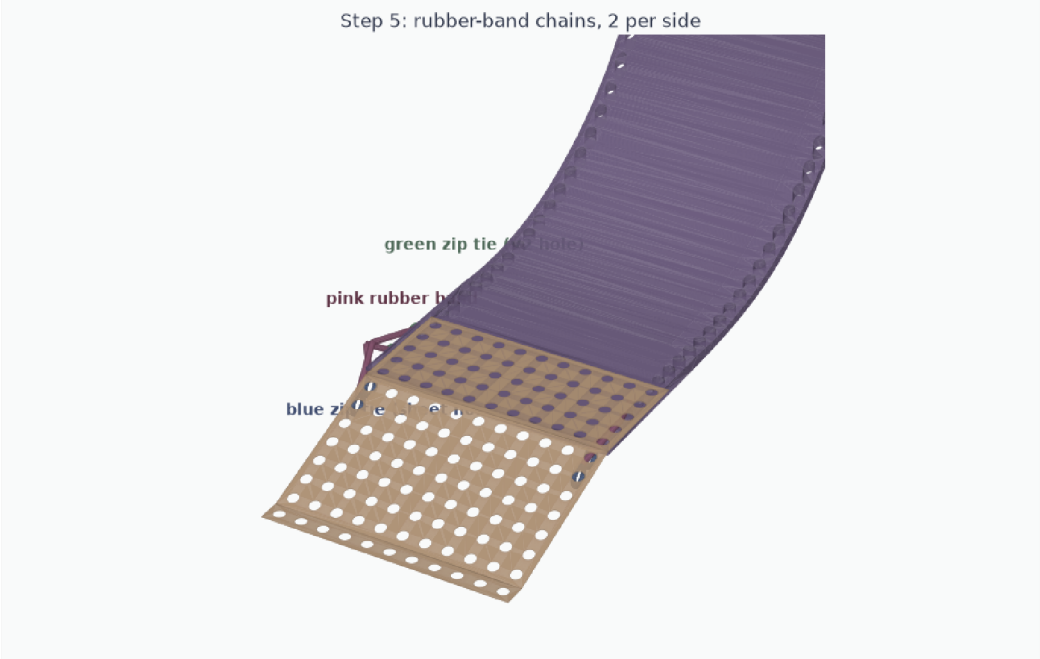


Side view (band pulls tongue into slot)



Tips

- **Do not over-tighten** the zip ties. Leave a **loose loop** so the band can stretch.
- Band tension should hold the tongue **fully seated** in the slot.
- Check that **all 4 bands match**. Replace any band that is cracked.
- V2 slot pitch is 7mm and the plate is 8mm, so **pick holes that line up**.
- Trim zip tie tails so they do not touch the ball.



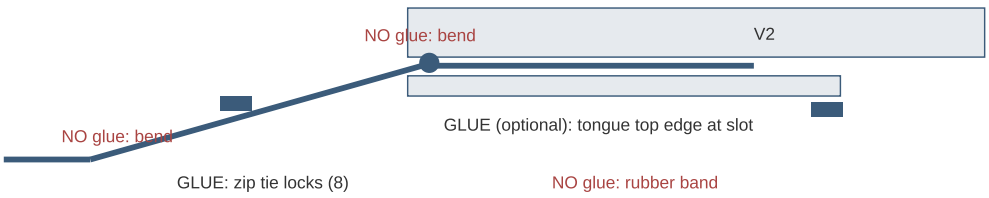
6 Super glue (thin CA)

Glue: a small drop on each of the 8 zip tie locks after tightening, so they do not slip. Optional: a thin line along the tongue's top edge where it meets the V2 slot or lip, so the tongue cannot pull out.

Do NOT glue: the two bend lines, the slope, the tail on the floor plate, or the rubber bands (CA makes rubber brittle). The bend must stay flexible.

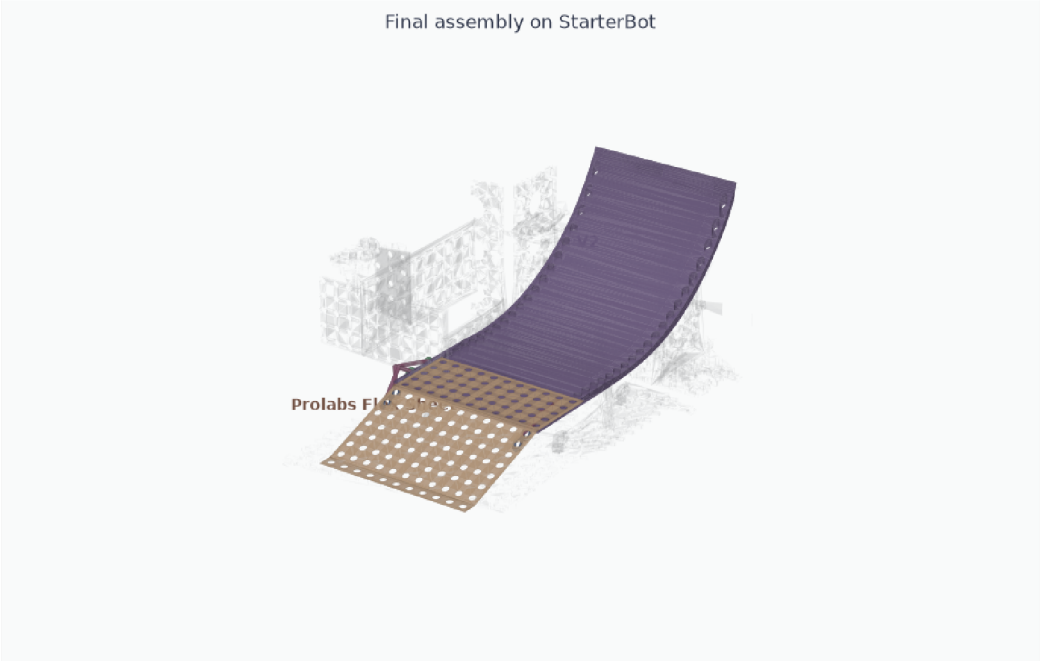
PETG note: CA sticks poorly to PETG. Lightly sand, wipe with IPA, then use a CA primer first. Cure: about 30 to 60 seconds to handle, 24 hours full strength. Do not run the robot before 1 hour.

Glue points (side view)



7 Final assembly

Mount V2 on the shooter. Check the tail rests on the floor plate and the sheet runs smoothly from floor to V2.



8 Hand check

Run the checklist below before using it on a robot.

6. Known issues (read before building)

1. **0.2mm sheet sags under the ball**
   A single 0.2mm layer will droop under a 71mm ball and will not hold a bend between tie points. The ball path shape is not controlled. We think a thicker sheet (0.6 to 0.8mm) or ribs would be much safer.

2. **Can tear at the holes**
   If the bands pull hard, the thin sheet can tear between holes. Use light bands and check after every run.

3. **Hole pitch 7mm vs 8mm**
   V2 holes at x +-40 are about 7.07mm apart (under the slot: y -32.5, -25.5, -18.4, -11.3, -4.2, measured in the STL). The sheet is on 8mm, so only some holes line up. Pick holes that line up for the chains.

4. **1.1mm step at V2 lip**
   The sheet sits in the slot under V2's top skin, so the ball meets a 1.1mm step up onto V2. Chamfer the V2 lip edge.

5. **Ball coasts undriven**
   About 131mm with nothing driving it, partly uphill. It needs about 0.7 m/s leaving the intake roller, or a push from the next ball.

6. **Kit part 368 in the path (CAD)**
   An upright plate at y -62 crosses the ball path in CAD. Check on the real robot.

7. **Not tested on a robot**
   Everything here is checked in CAD only.

7. Hand-check checklist

- [ ] V2 size within **+-0.5mm** of 129.35 x 148.57 x 88mm
- [ ] Sheet is one clean layer, no tears at holes
- [ ] Tongue goes fully 40mm into the slot
- [ ] Every zip tie lock has a glue drop and does not slip
- [ ] No glue on the two bend lines or on the band
- [ ] All 4 chains fitted, 2 per side, bands match
- [ ] Pull the sheet gently: it springs back into the slot
- [ ] Roll the 71mm ball by hand: watch how much the sheet sags
- [ ] V2 lip chamfered, no catch
- [ ] No clash with kit part 368 near y -62

8. V2 + Flex Sheet vs V5b

	V2 + Prolabs Flex Sheet	V5b
Parts	2 prints + chains	**1 print**
Material	55 cm3 (about 70g) + 1.9g sheet	227 cm3
Squeeze	14.5%	12.6%
Ramp shape control	Low (0.2mm sags, hand bent)	**Fixed shape**
Team-to-team variation	Higher	**Low**

Honest note: for untested manufacture for many teams, we think V5b is the lower-risk choice. V2 + Flex Sheet is light and cheap to try.

by Aarav Jaisingh, Prolabs Robotics. Feedback from teams is very welcome.